

Surface Brightness Fluctuations

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As a simple demonstration of the basic idea, consider the following. Take a galaxy at distance D and assume it is composed of identical stars of luminosity L . In projection, some small patch of the galaxy has mean intrinsic surface brightness I (in units of $L_\odot \text{ pc}^{-2}$). The corresponding mean number of stars per unit area is I/L . Suppose we have an image of the galaxy made up of square pixels that subtend a solid angle $\theta \times \theta$. Since the physical area subtended by a pixel is $(\theta D)^2$, the mean flux per pixel in the region with mean brightness I is

$$\langle f \rangle = I(\theta D)^2 / (4\pi D^2) = I\theta^2 / (4\pi) , \quad (1)$$

independent of distance. Of course, not every pixel has the mean flux. Because we are assuming that all stars have the same luminosity, flux variations from pixel to pixel result from differences in the number of stars from the mean value

$$\langle n \rangle = I(\theta D)^2 / L , \quad (2)$$

so that $\langle f \rangle = \langle n \rangle L / (4\pi D^2)$, as must be the case. From the Poisson distribution, the standard deviation in the number of stars from pixel to pixel is $\sqrt{\langle n \rangle}$ (see the formal presentation below), so that the flux variation is

$$\sigma_f = \sqrt{\langle n \rangle} L / (4\pi D^2) = \langle f \rangle / \sqrt{\langle n \rangle} . \quad (3)$$

It follows that

$$\sigma_f^2 / \langle f \rangle = L / (4\pi D^2) , \quad (4)$$

the simple, final result. Here the stellar luminosity L acts like a standard candle. Given the known L and the measured values of σ_f and f , the quantity σ_f^2 / f equals the flux per star, from which we obtain the distance. Note also that the fractional fluctuations are $\sigma_f / f = 1 / \sqrt{\langle n \rangle} \propto D^{-1}$; that is, if the galaxy is placed at a larger distance, its image will appear smoother. We now generalize these results to incorporate a distribution of stellar luminosities. Before we get to that, let's digress a bit.

What is the value of $\langle n \rangle$ that we should expect? This can be crudely estimated as follows. Elliptical galaxies tend to have a surface brightness that falls off smoothly from the center of the galaxy. Typical total luminosities and characteristic radii are $L_g \sim 10^{10} L_\odot$ and $R \sim 5 \text{ kpc}$. This radius contains roughly half of the total light. Thus, the characteristic surface brightness is $L_g / (\pi R^2) = \mathcal{O}(10^2 L_\odot \text{ pc}^{-2})$. Given a distance D and pixel size θ , we find from eq. (2) that

$$\langle n \rangle \sim 10^3 [\theta('') D(\text{Mpc})]^2 \quad (5)$$

for a typical stellar luminosity of $\sim L_\odot$. So, we see that generally $\langle n \rangle \gg 1$. At the $\approx 20 \text{ Mpc}$ distance of the Virgo cluster, we find $\langle n \rangle \approx 4 \times 10^5$, such that $1 / \sqrt{\langle n \rangle} \sim 10^{-3}$, which is smaller than the photometric precision that is typically attainable from the ground. However, the actual situation is better than this when we consider a more realistic stellar luminosity function.

When the stellar luminosities are drawn from a broad distribution that is known a priori, what is the value of the characteristic luminosity $\bar{L}$ defined by

$$\bar{L} \equiv 4\pi D^2 \sigma_f^2 / f ? \quad (6)$$

We again consider a pixelated image of a galaxy and focus on a relatively small neighborhood about a point where the mean surface brightness is I . Suppose this neighborhood is defined by an array of N_p pixels, each of which is labeled by an index k . Each pixel contains n_k stars, such that $\sum_k^{N_p} n_k = N_s$, the total number of stars in the array. Therefore, the mean number of stars per pixel is $\langle n \rangle = N_s/N_p$. We assume that the stars in each pixel have a spectrum of luminosities drawn from the same distribution. Now, imagine binning stars of similar luminosities in such a way that there are $n_k(i)$ stars of luminosity in the range $(L_i - \Delta L_i/2, L_i + \Delta L_i/2)$, where ΔL_i is some small, but otherwise arbitrary bin width. It follows that the total flux from pixel k is

$$f_k = \frac{1}{4\pi D^2} \sum_i n_k(i) L_i . \quad (7)$$

Averaging over the whole array, we obtain the mean flux per pixel:

$$4\pi D^2 \langle f \rangle = \frac{1}{N_p} \sum_k \sum_i n_k(i) L_i = \sum_i L_i \left[\frac{1}{N_p} \sum_k n_k(i) \right] = \sum_i L_i \langle n(i) \rangle , \quad (8)$$

where $\langle n(i) \rangle$ is the expectation for the number of stars in luminosity bin i from the underlying luminosity function; this assume that N_s is sufficiently large that the luminosity function is very well sampled over the full range. Thus, we have $4\pi D^2 \langle f \rangle = \langle n \rangle \langle L \rangle$, the mean stellar luminosity. The pixel-to-pixel variance of the flux is defined by

$$\sigma_f^2 = \langle [f - \langle f \rangle]^2 \rangle = \langle f^2 \rangle - \langle f \rangle^2 . \quad (9)$$

The mean-square flux is given by

$$\begin{aligned} (4\pi D^2)^2 \langle f^2 \rangle &= (4\pi D^2)^2 \frac{1}{N_p} \sum_k f_k^2 = \frac{1}{N_p} \sum_k \left[\sum_i n_k(i) L_i \right]^2 \\ &= \frac{1}{N_p} \sum_k \left[\sum_i n_k^2(i) L_i^2 + \sum_i \sum_{j \neq i} n_k(i) n_k(j) L_i L_j \right] \\ &= \sum_i L_i^2 \langle n^2(i) \rangle + \sum_i \sum_{j \neq i} L_i L_j \langle n(i) n(j) \rangle , \end{aligned} \quad (10)$$

Similarly, note that

$$(4\pi D^2)^2 \langle f \rangle^2 = \sum_i L_i^2 \langle n(i) \rangle^2 + \sum_i \sum_{j \neq i} L_i L_j \langle n(i) \rangle \langle n(j) \rangle . \quad (11)$$

There is no reason to think that there would be any correlation between different luminosity bins, and so we have $\langle n(i) n(j) \rangle = \langle n(i) \rangle \langle n(j) \rangle$. Therefore, the cross terms cancel in the difference $\langle f^2 \rangle - \langle f \rangle^2$, and we are left with

$$(4\pi D^2)^2 \sigma_f^2 = \sum_i L_i^2 (\langle n(i)^2 \rangle - \langle n(i) \rangle^2) = \sum_i L_i^2 \langle n(i) \rangle , \quad (12)$$

where the last step assumes that each of the $n_k(i)$ are drawn from the Poisson distribution with variance $\langle n(i) \rangle$. Finally, we have the fundamental result for surface brightness fluctuations:

$$4\pi D^2 \sigma_f^2 / \langle f \rangle \equiv \bar{L} = \langle L^2 \rangle / \langle L \rangle . \quad (13)$$

In addition, we find (compare to eq. [3])

$$\sigma_f = \sqrt{\langle n \rangle} \sqrt{\langle L^2 \rangle} / (4\pi D^2) , \quad (14)$$

so that the fractional standard deviation is

$$\sigma_f / \langle f \rangle = \sqrt{\langle L^2 \rangle} / (\sqrt{\langle n \rangle} \langle L \rangle) . \quad (15)$$

As we will see, for a realistic luminosity function, $\sqrt{\langle L^2 \rangle} \gg \langle L \rangle$ and $\sigma_f / \langle f \rangle \gg 1 / \sqrt{\langle n \rangle}$.

Since we are mainly interested in elliptical galaxies, we assume for simplicity that all the stars have the same age of ~ 10 Gyr. In other words, we assume that the turn-off mass is $\simeq 1 M_\odot$. For many galaxies, this isn't an unreasonable starting point. Here we separately treat low-luminosity end of the luminosity function (LF), populated primarily by main-sequence stars, and the high-luminosity end that includes evolved giants.

The shape of the main-sequence branch of the LF is determined by the stellar initial mass function, we assume has the form

$$p(m) \propto m^{-2.5} , \quad (16)$$

and the mass-luminosity relation,

$$L(m) / L_\odot \simeq (m / M_\odot)^4 . \quad (17)$$

Therefore, the luminosity function has the approximate power-law form

$$p(L) = p(m) |dm/dL| \propto L^{-1.4} . \quad (18)$$

Because this calculation is restricted to main-sequence stars, the maximum luminosity for this branch is $L(m_{\text{to}})$, where $m_{\text{to}} \simeq 1 M_\odot$ is the turn-off mass.

Evolution on the red giant branch is very well described by the well known relation between core mass and luminosity. This relation is usually written as a simple power law,

$$L \propto \dot{m}_c \propto m_c^\alpha , \quad (19)$$

where $\alpha = 6 - 8$, depending largely on the metallicity. Note that the core mass and luminosity both increase more or less monotonically. For a stellar population that has a single age at the current epoch, and given that the lifetime of the giant phase is only $\sim 10\%$ of the main-sequence lifetime, stars on the giant branch have essentially the same mass. The probability of finding a giant with luminosity in the range $(L, L + dL)$ is proportional to the time any single star spend in the range, given by

$$p(L) \propto dt \propto dL / \dot{L} . \quad (20)$$

From the $L(m_c)$ relation, we have

$$\dot{L} \propto m_c^{\alpha-1} \dot{m}_c \propto L^{2-1/\alpha} , \quad (21)$$

and thus

$$p(L) \propto L^{-2+1/\alpha} . \quad (22)$$

The limits of the LF in this range are $\sim 10L_\odot$ at the lower end and a few $\times 10^3 L_\odot$ at the upper end.

One conclusion we draw from the above analysis—insofar as our assumption are valid—is that the power-law slope of the LF is everywhere in the range -1 to -2 . This is a critical point when trying to evaluate the fluctuation luminosity $\bar{L} \equiv \langle L^2 \rangle / \langle L \rangle$. Let us assume that the LF everywhere has the form $p(L) = AL^{-\beta}$, where $\beta = 1-2$. The first moment are then given by

$$\langle L \rangle = \begin{cases} A(L_{\max} - L_{\min}) , & \beta = 1 , \\ A \ln(L_{\max}/L_{\min}) , & \beta = 2 . \end{cases} \quad (23)$$

For the second moment, we have

$$\langle L^2 \rangle = \begin{cases} A(L_{\max}^2 - L_{\min}^2)/2 , & \beta = 1 , \\ A(L_{\max} - L_{\min}) , & \beta = 2 . \end{cases} \quad (24)$$

When $L_{\max}/L_{\min} \gg 1$, we can neglect differences involving $L_{\min}$, and we obtain

$$\bar{L} = \begin{cases} L_{\max}/2 , & \beta = 1 , \\ L_{\max}/\ln(L_{\max}/L_{\min}) , & \beta = 2 . \end{cases} \quad (25)$$

If $L_{\max}/L_{\min} \sim 10^6$, then $\bar{L} \sim (0.1-0.5) \times L_{\max} \sim 10^2-10^3 L_\odot$.

In order to estimate the actual value of $\langle L \rangle$ we need the normalization A:

$$A \simeq \begin{cases} 1/\ln(L_{\max}/L_{\min}) , & \beta = 1 , \\ L_{\min} , & \beta = 2 . \end{cases} \quad (26)$$

Therefore,

$$\langle L \rangle \simeq \begin{cases} L_{\max}/\ln(L_{\min}/L_{\max}) , & \beta = 1 , \\ L_{\min} \ln(L_{\min}/L_{\max}) , & \beta = 2 . \end{cases} \quad (27)$$

which varies between $\sim 10^2 L_\odot$ and $\sim 10^{-2} L_\odot$ as β increases from 1 to 2. We then find that $\bar{L}/\langle L \rangle$ ranges from ~ 10 ($\beta = 1$) to $\sim 10^4$ ($\beta = 2$), and we expect $\sigma_f/\langle f \rangle \sim 10-100$ times $1/\langle n \rangle$. Therefore, $\sigma_f/\langle f \rangle$ may be as large as $\sim 10^{-3}(100 \text{ Mpc}/D)^2$ for an angular resolution of $1''$.